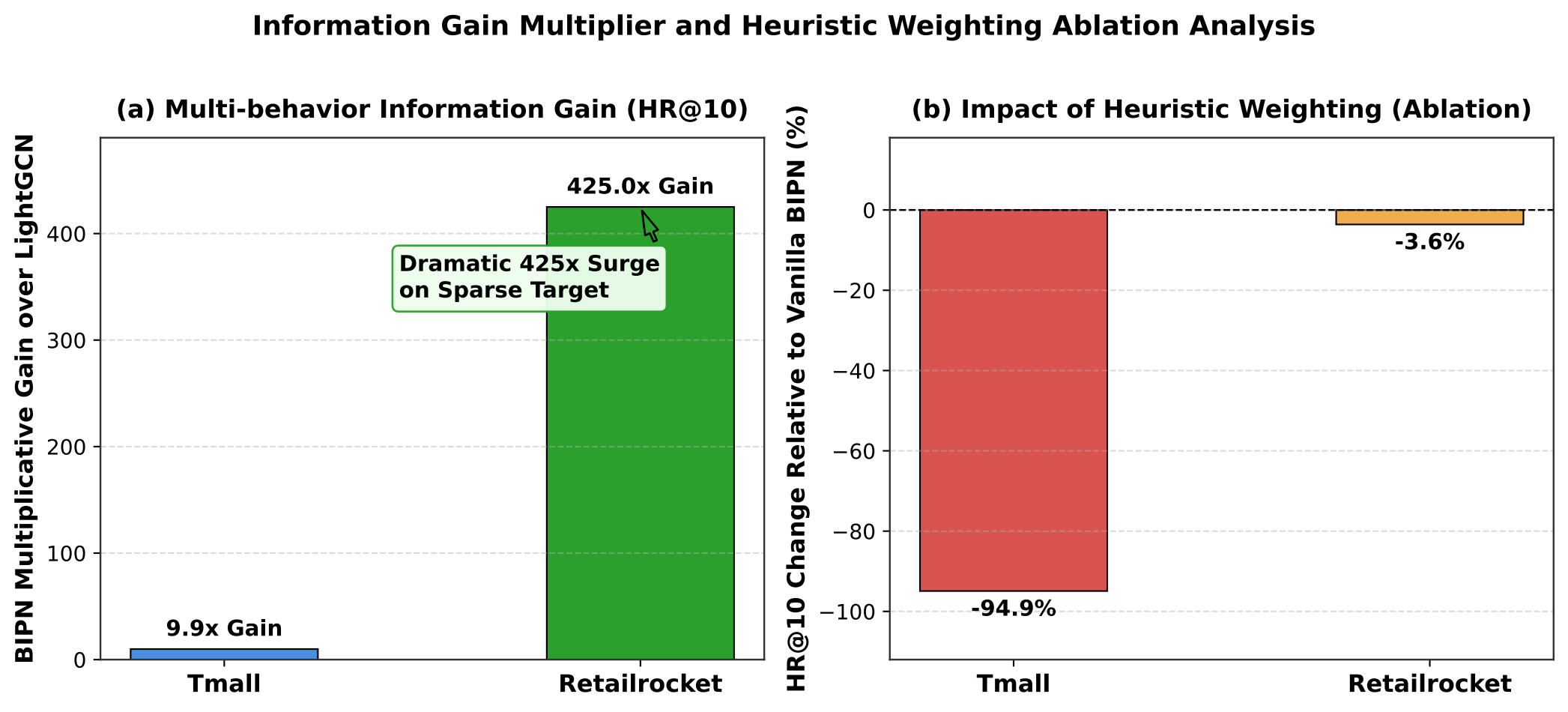
Information Gain Multiplier and Heuristic Weighting Ablation Analysis
(a) Multi-behavior Information Gain (HR@10)
BIPN Multiplicative Gain over LightGCN
0
100
200
300
400
9.9x Gain
425.0x Gain
Dramatic 425x Surge
on Sparse Target
Tmall
Retailrocket
(b) Impact of Heuristic Weighting (Ablation)
HR@10 Change Relative to Vanilla BIPN (%)
0
−20
−40
−60
−80
−100
-94.9%
-3.6%
Tmall
Retailrocket